The spectral square root of Q_L

high-pass on the evaluators, crush on the desert

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Abstract

$Q^{1/2} = \sum \sqrt{\lambda_j}\, v_j v_j^\top$ is the unique PSD factor. Its top eigenspace is the span of the in-band evaluators $\hat{\eta}(\gamma_k)$ (1–1 pairing for $K \leq 6$). Its bottom eigenspace is the radical: $\sqrt{\lambda_{\min}} \sim 10^{-24}$ at $\mu = 11$. As an operator on V_N, $Q^{1/2}$ is a high-pass tuned to the zeros, not a list of Euler lags. Polar of the evaluator matrix C recovers $Q^{1/2}$ only after the off-band tail is in; with four in-band zeros on V_{13} the gap is 47%. No new closed form.

1 The operator

On V_N,

$$Q^{1/2} f = \sum_{j=1}^{N} \sqrt{\lambda_j}\, \langle f, v_j \rangle\, v_j.$$

If C is the matrix with columns $\hat{\eta}(\gamma_k)$ then the explicit formula says $Q \approx CC^\top$. The left polar factor of C is $(CC^\top)^{1/2}$. Hence

$$Q^{1/2} \;\approx\; (CC^\top)^{1/2}$$

as soon as the tail of γ beyond $\omega_{\max}$ is small. At $\mu = 11$, $\|Q - Q_K\|_F / \|Q\|_F$ is 8.9% at $K = 40$ ($\gamma \approx \omega_{\max}$) and 5% at $K = 60$ (notebook P1). The same tail sits in the polar: $Q^{1/2}$ and $(CC^\top)^{1/2}$ differ by that remainder, not by a new operator.

2 Two regimes

High. Greedy pairing of the largest v_j with the in-band $\hat{\eta}(\gamma_k)$ is 1–1 for $K \leq 6$ (mean cosine 0.81 at $N = 13$, $K = 4$). It breaks at the last in-band zero ($K = 7$, min cosine 0.27). Those $\sqrt{\lambda_j}$ are $O(1)$ (on V_{13}: $1.74, 1.59, 1.56, 1.42$). $Q^{1/2}$ acts on that subspace as a mild gain.

Low. The remaining eigenvalues follow the Slepian staircase of the desert and then the razor. On V_{13} already $\sqrt{\lambda} \approx 0.27, 0.024, 0.001, \ldots$. On V_{47}, $\sqrt{\lambda_{\min}} \approx 6 \times 10^{-24}$. $Q^{1/2}$ crushes the ground state. That is the same hyper-nullity as MUSIC: the radical is the approximate kernel of every in-band evaluator, hence of $C^\top$, hence of $Q^{1/2}$.

3 What this is not

$Q^{1/2}$ is not an arithmetic T_L. Its rows in the cosine basis are the v_j weighted by $\sqrt{\lambda_j}$, and the v_j of the top are linear combinations of evaluators at the γ_k, not at the $\log p$. Passing from $Q^{1/2}$ to a factor whose rows are lags $\tau_{\log p}$ is the SOS problem already recorded as open (*sos-arithmetic*). The spectral square root is the unique PSD answer to "write Q as a norm"; it answers that question by pointing back at the zeros.

Status

Analyzed. $Q^{1/2}$ is the polar of the evaluator synthesis, up to the off-band tail. High-pass on $\mathrm{span}\{\hat{\eta}(\gamma_k)\}$, crush on the desert. Not a closed arithmetic factor. Not RH.